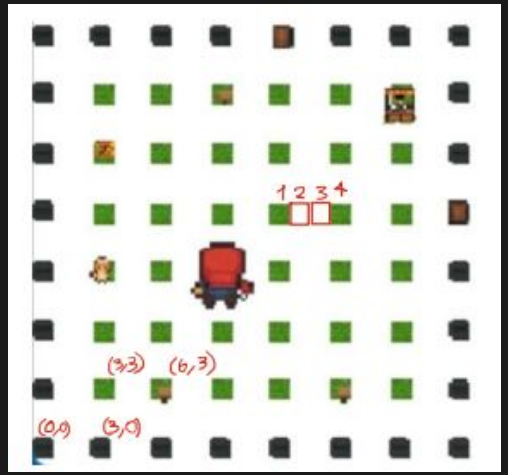


CaveMon

Programación II - ETSIINF UPM - 26/05/2025

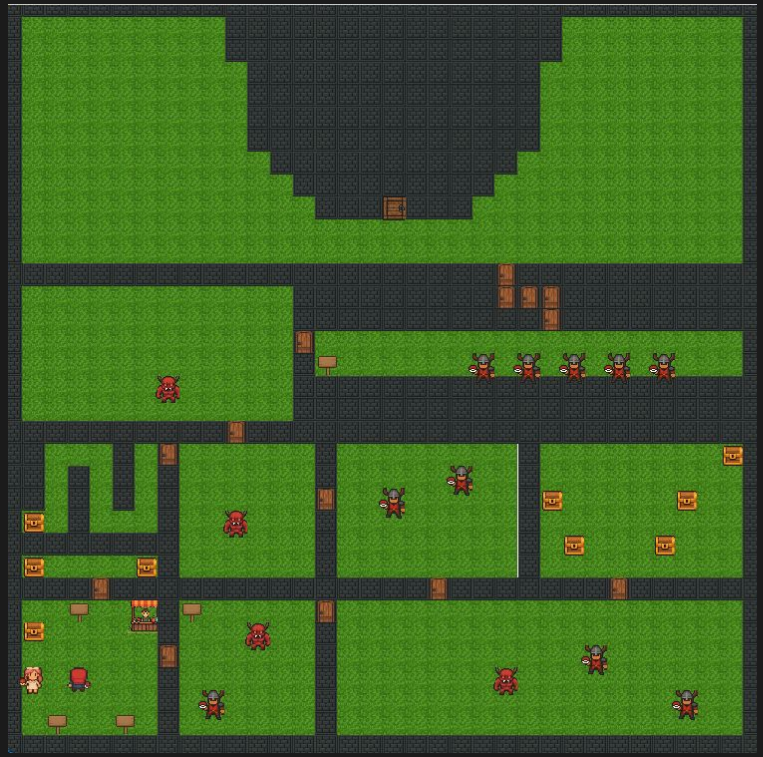
Diego Gutiérrez & Marcos Almonacid

Mapa



1	1	1	1	1	1	1	1
1	0	0	4	0	0	7	1
1	9	0	0	0	0	0	1
1	0	0	0	0	0	0	2
1	6	0	0	0	0	0	1
1	0	0	0	0	0	0	1
1	0	4	0	0	4	0	1
1	1	1	1	1	1	1	1

LEVEL DESIGN



#RULES

RULES

0: Suelo

1: Muro

2: Borde Menu

3: Relleno Menu

4: NPC

5: Chest

6: Inventory Background

7: Button

8: Mon Chest

9: Healer

10: Enemigo

11: Tienda

12: Puerta Cerrada

13: Puerta Abierta

14: dibujar nada, Vacio

15: WildMon

```
1 1 1 1 1 1 1 1
1 0 0 4 0 0 7 1
1 9 0 0 0 0 0 1
1 0 0 0 0 0 0 2
1 6 0 0 0 0 0 1
1 0 0 0 0 0 0 1
1 0 4 0 0 4 0 1
1 1 1 1 1 1 1 1
```

w: arriba

s: abajo

d: derecha

a: izquierda

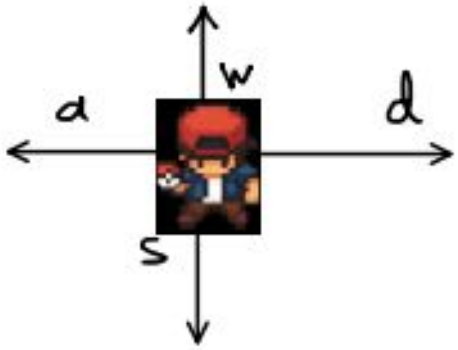
e: interactuar con NPC

m: No collision cheat

i: Print inventory

ñ: Cheat Coins

Movimiento



(dx, dy)

$w = (0, 1)$

$s = (0, -1)$

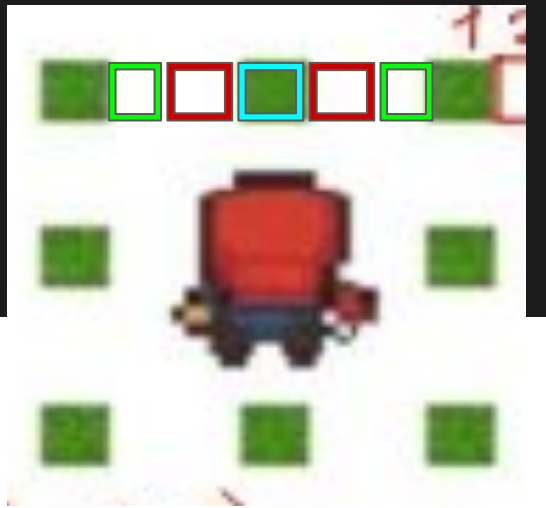
$d = (1, 0)$

$a = (0, -1)$

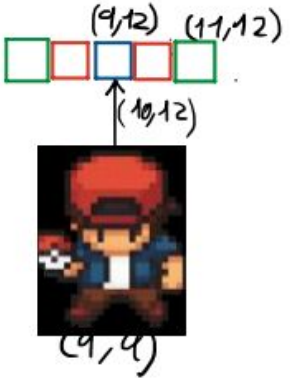
Looking



Colisiones Lógica



Caso 'W':




```
collisionNumbers.addAll(Arrays.asList(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 15, 16));
```

$(dx, dy) = (0, 1)$

mapa.getVal$(x + 3 \cdot dx, y + 3dy) = (9+0, 9+3\cdot 1) = (9, 12)$

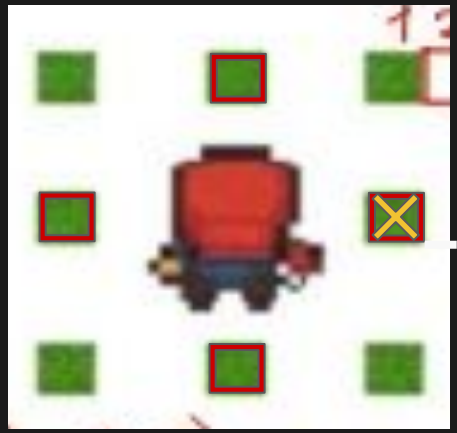
" $(x + 3dx \pm 1dy, y + 3dy \pm dx) = (9+0\pm 1, 9+3\pm 0) = (10, 12)$

" $(x + 3dx \pm 2dy, y + 3dy \pm 2dx) = (11, 12)$

□ || □ || □ ∈ collisionNumbers.

interact()

LinkedList<NPC>



TRUE:
pos = (x+3,y)



(12, 15)

(30, 30)



(12, 15)

```
for(int i = 0; i < npcs.size(); i++){
    if(npcs.get(i).posX() == posNpc[0] && npcs.get(i).posY() == posNpc[1]){
        npcInteres = npcs.get(i);
        System.out.println(x:"Npc interactuado");
    }
}
```

Enemy & Battle



Enemy AI





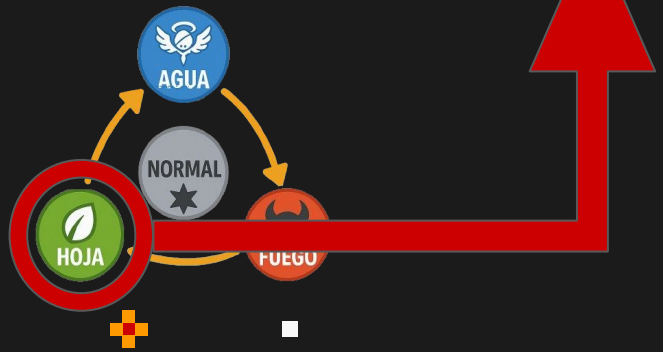
Ola Rompedora
Agua

Látigo Enredado
Hoja





whatCrits()



Random + ArrayList()

Positive:

Ola Rompedora
Agua

Látigo Enredado
Hoja

Default

Ola Rompedora
Agua

Ola Rompedora
Agua

Player

ArrayList<Mon>()

get(i)



- changeMon()
- sideMenu()
- ...







0 1 2

Enemy

LinkedQueue<Mon>()



- heal()
- allDead()
- nextMon()

poll()

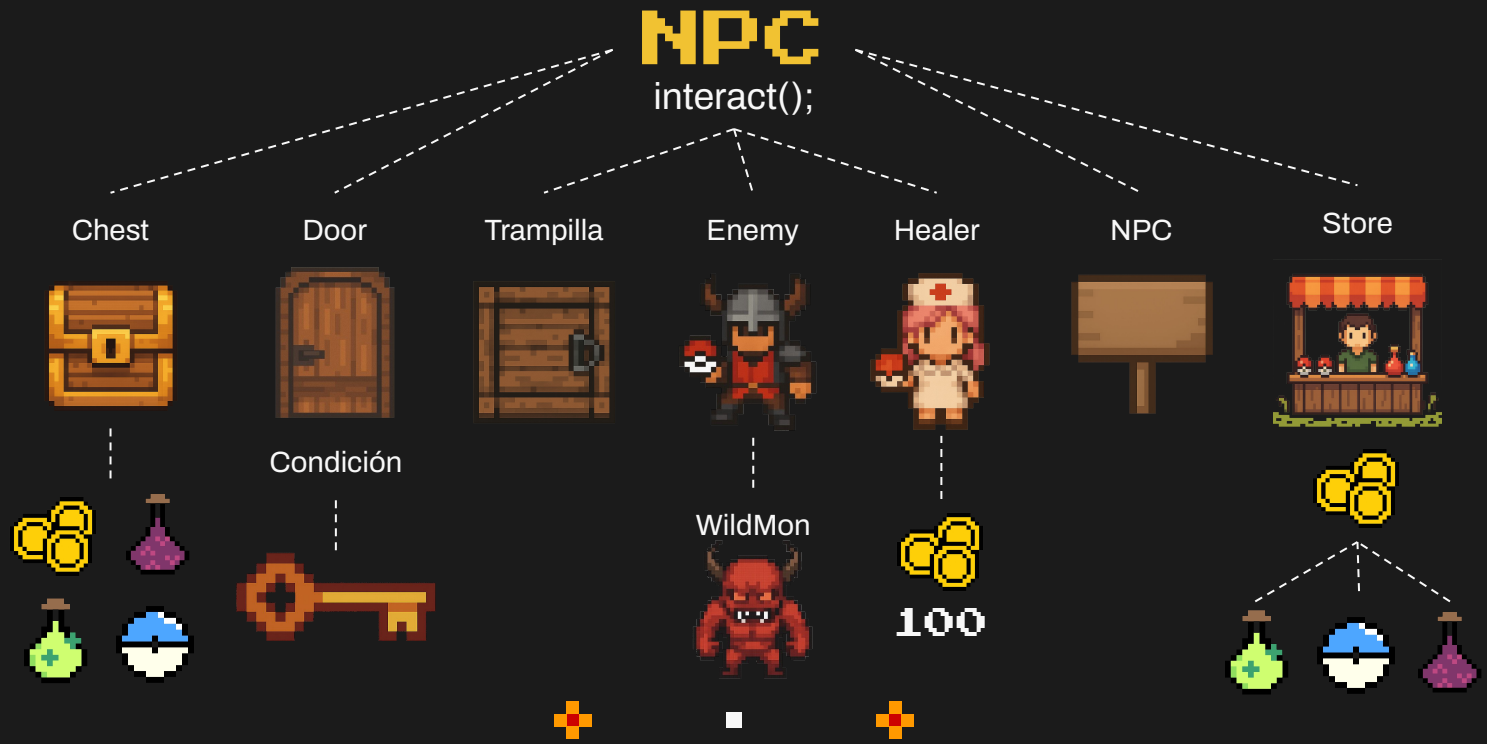






peek() peek()

Clases Abstractas



Clases Abstractas

Items

use();

Coins



Health Potion



10% MAXHP

Strength Potion

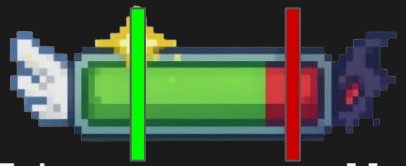


x2 DMG

MonBall



Rand(MAXHP) > hp*Mod



Si

No

Clases Abstractas

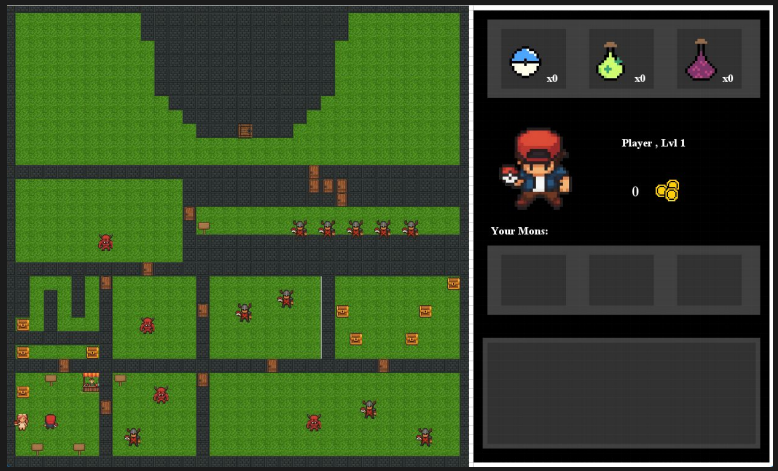
Map

initializeMenu();
initializeWalls();

Battle



Level

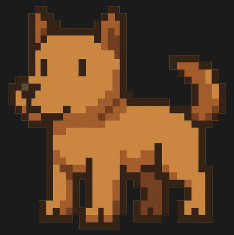


Interfaz

IMon

getMon();

PerroMon



NeuroMon



AquaMon

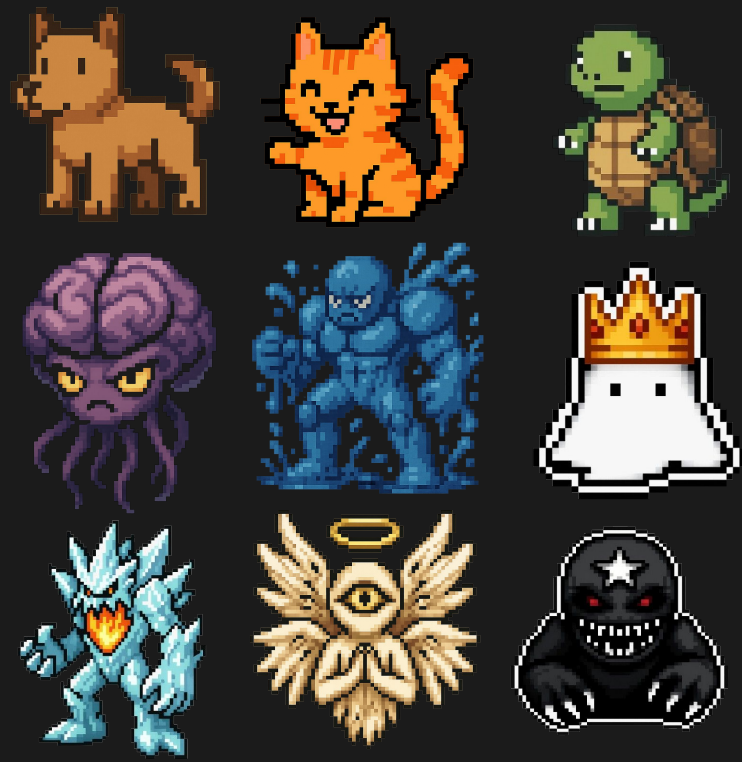


AngelMon



Balance

Nombre	Tipo	HP base	DMG base
TortugaMon	Normal	1500	1,3
AquaMon	Agua	1200	1,6
GlitchMon	Fuego	1000	1,8
KingMon	Hoja	1400	1,4
NeuroMon	Normal	1300	1,5
GatoMon	Hoja	1100	1,7
PerroMon	Fuego	1350	1,3
IceHotMon	Agua	1400	1,35
AngelMon	Normal (Bo	1650	1,6
PesadillaMon	Fuego (Bos	1600	1,7

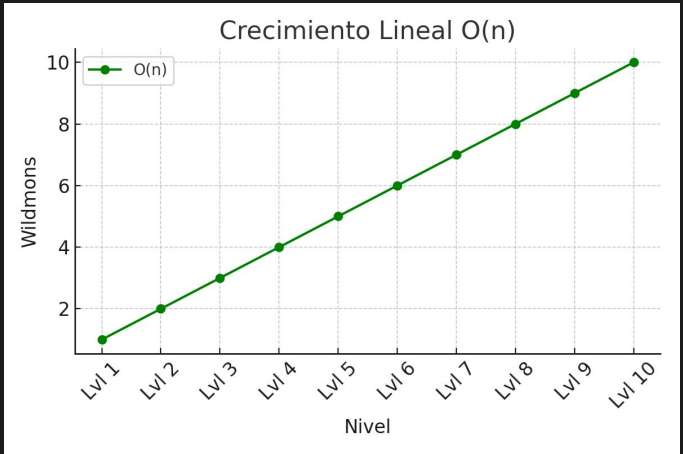


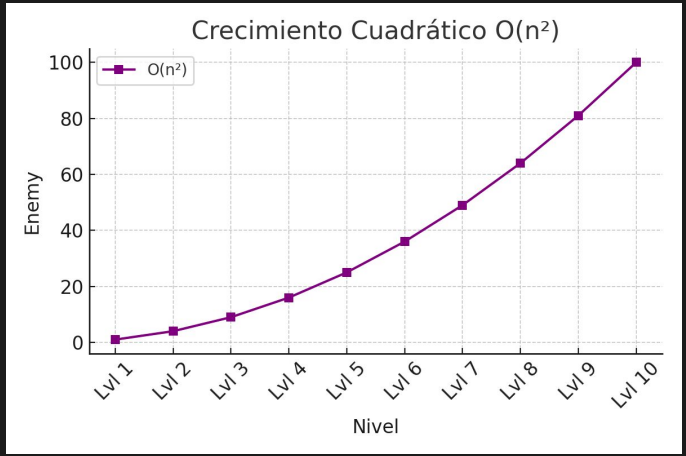
YOU WIN!!!

Escalado Infinito

WildMons O(n)

Enemies y MonChests O(n^2)






x0
x0
x0
Player , Lvl 1
0
our Mons:
END
Gracias

Esto por si te hace falt

